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Version No.				
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MATHEMATICS HSSC-I
(Old Curriculum)
SECTION – A (Marks 20)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نگران مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پُر کریں۔ **Invigilator Sign.** _____

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** _____

	Question	A	B	C	D	A	B	C	D
1.	If order$(A) = m \times n$, order$(B) = n \times p$ then order $(AB)' =$	$m \times m$	$p \times m$	$p \times n$	$n \times m$	○	○	○	○
2.	If $Z = 2i^2 - 3i$ then $img(Z) =$	–1	–3	2	3	○	○	○	○
3.	The rank of matrix: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$	0	1	2	3	○	○	○	○
4.	$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} =$	0	15	12	7	○	○	○	○
5.	For what value of P, $[2, P, 5]$ is orthogonal to $[3, 1, P]$?	–2	–3	–5	–1	○	○	○	○
6.	The work done in moving an object along a vector $9\hat{i} - \hat{j} + \hat{k}$ if applied force is $3\hat{i} + 2\hat{j} + \hat{k}$ is:	24 unit	26 unit	28 unit	30 unit	○	○	○	○
7.	A sequence $\{a_n\}$ is an arithmetic sequence if $\forall n \in N$ and $n > 1$	$\frac{a_n}{a_{n-1}} = d$	$a_{n+1} - a_{n-1} = d$	$\frac{a_{n-1}}{a_n} = d$	$a_n - a_{n-1} = d$	○	○	○	○
8.	Find: 2+6+18+.....6 terms	827	726	728	627	○	○	○	○
9.	$\sum_{j=1}^{n} j^3 =$	$\frac{n(n+1)}{2}$	$\left(\frac{n(n+1)}{2}\right)^2$	$\frac{n(n+1)(2n+1)}{6}$	$\left(\frac{n(n-1)}{2}\right)^2$	○	○	○	○
10.	The value of $\frac{3!6!}{7!5!}$ is	$\frac{1}{7!}$	$\frac{1}{20!}$	20!	$\frac{1}{140}$	○	○	○	○
11.	If $P(A \cap B) = \frac{3}{9}$, $P(B) = \frac{4}{9}$ then $P(A/B) =$ is:	$\frac{3}{4}$	$\frac{4}{3}$	$\frac{3}{9}$	$\frac{4}{9}$	○	○	○	○

	Question	A	B	C	D	A	B	C	D
12.	If 'n' is even then the middle term in the expansion of $(a+b)^{2n}$ is:	$(2n+2)^{th}$ *term*	$(2n+1)^{th}$ *term*	$(n+1)^{th}$ *term*	$\left(\frac{2n+1}{2}\right)^{th}$ *term*	○	○	○	○
13.	Number of terms in the expansion of $(1+2x)^{\frac{2}{3}}$ is:	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{6}$	Infinite	○	○	○	○
14.	Which of the following is **NOT** a solution of inequality? $3x-2y\leq 15$	$(0,0)$	$(3,1)$	$(-3,-2)$	$(9,0)$	○	○	○	○
15.	$\text{Cos}\,3\alpha =$	$4\text{Cos}^3\alpha - 3\text{Cos}\,\alpha$	$3\text{Cos}\,\alpha - 4\text{Cos}^3\alpha$	$4\text{Cos}\,\alpha - 3\text{Cos}^3\alpha$	$3\text{Cos}^3\alpha - 4\text{Cos}^2\alpha$	○	○	○	○
16.	If $\text{Sin}\,\theta = \frac{-1}{2}$ and θ does not lie in 3rd quadrant then value of $\tan\theta$ lies in:	1st quadrant	2nd quadrant	3rd quadrant	4th quadrant	○	○	○	○
17.	If a=3, b=9, c=7 are sides of triangle then smallest angle is:	α	β	γ	Both β *and* γ	○	○	○	○
18.	The period of $5\text{Sin}\,3x$ is:	2π	3π	$\frac{2\pi}{3}$	$\frac{3\pi}{2}$	○	○	○	○
19.	The maximum value of $y=1+3\text{Sin}(2x-15)$ is:	$\frac{1}{4}$	$\frac{-1}{2}$	4	–2	○	○	○	○
20.	If $\text{Cos}^{-1}\frac{1}{2}=\frac{\pi}{2}-X$ then $X=$	90°	60°	30°	45°	○	○	○	○

——1HA-I 25008- 90081(OLD) ——

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